

Final Exam - EEE3094S 2020 - MEMO

Information

- The exam is open book.
 - There are six questions, totalling 100 marks. You must answer all of them.
 - Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint. A .pptx file containing the plots has been provided.
 - You must explain your reasoning and show your calculations for all questions.
 - If you create or modify a plot to support your answer, ensure that you reference this graphic in your answer. You should also label the graphic with the question number. It is also helpful to annotate important features to draw attention to them.
 - Closed-loop systems are in unity feedback unless otherwise stated.
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Question 1

Consider the following block diagrams:

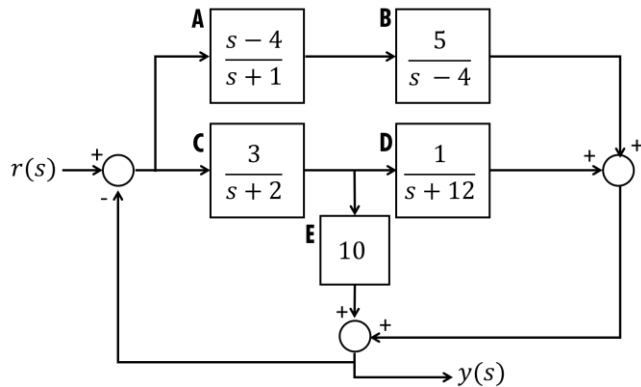


Figure 1.A - Block diagram A

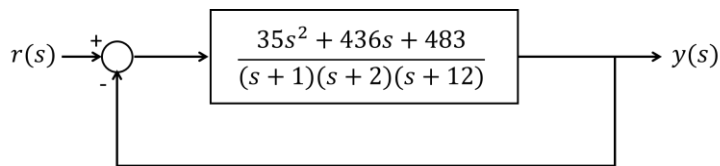


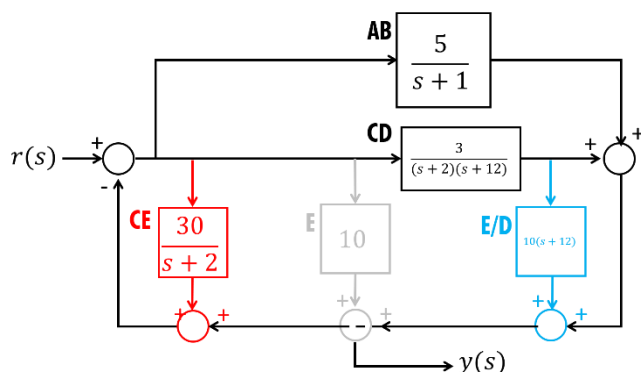
Figure 1.B - Block diagram B

A) [4 marks] Show that the two diagrams are equivalent.

Solution:

Possible steps:

- Combine A and B into AB
- Move the summing junction after E.
 - Option 1: before C. C must be added to branch, so C and E combine into CE.
 - Option 2: after D. $1/D$ must be added to branch, so $1/D$ and E combine into E/D . The resulting parallel path can be simplified to $1 + E/D$.
- Combine C and D into CD. If Option 2 was followed, combine with $1 + E/D$ to give $CD + CE$.
- Combine the three parallel branches into $AB + CD + CE$ and simplify to get the transfer function of the block in diagram B.



B) [2 marks] Reduce the open-loop transfer function in the second diagram to a second-order model.

Solution: The pole at -12 is much faster than the other two, so it can be replaced with a gain of $1/12$. The reduced order model is $\frac{35s^2 + 436s + 483}{(s+1)(s+2)12}$. Optional: the zeros are at -11.2 and -1.2. You may remove the zero at -11.2 and replace it with a gain of 11.2.

C) [2 marks] Subsystem B in the first block diagram is unstable. Do you trust the stability of the overall system? Explain.

Solution: No – the zero in A theoretically cancels the unstable pole in B, but if either of these models is inaccurate, perfect cancellation will not be achieved and the combined system will still be unstable.

Consider the following system:

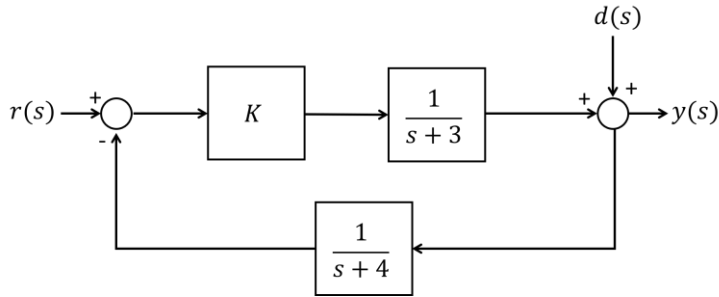


Figure 1.C – Block diagram C

D) [3 marks] Taking the gain $K = 1$, Find the transfer function relating the disturbance to the output of the system, and calculate its final value for a disturbance that can be modelled as $d(t) = 5u(t)$.

Solution: The disturbance is related to the output by the transfer function $y(s) = \frac{1}{1+Q(s)H(s)} d(s)$, where $Q(s)$ is the combined forward path and $H(s)$ is the feedback path.

The final value of the error for the given input is $\lim_{s \rightarrow 0} s \frac{1}{1+Q(s)H(s)} \frac{5}{s} = \frac{5}{1 + \lim_{s \rightarrow 0} Q(s)H(s)} = \frac{5}{1 + (0.33)(0.25)} = 4.62$

E) [2 marks] Calculate the gain K required to reduce the effect of the disturbance to 10% of the value you worked out in (D).

Solution: The disturbance can be decreased by increasing the gain K .

$$\frac{5}{1 + K(0.33)(0.25)} \leq 0.462 \therefore K(0.33)(0.25) \geq \frac{5}{0.462} - 1 \therefore K \geq 118$$

F) [2 marks] With reference to the type number of the system, explain how you could reduce the effect of the disturbance to zero.

Solution: This is a type 0 system, meaning it has no integrators (i.e. poles at 0) in the loop path. Increasing the type number to 1 by adding an integrator to the controller ($\frac{K}{s}$) would effectively give infinite DC gain, and reduce the error constant to zero.

Total: 15 marks

Question 2

A closed-loop system and its root locus are shown below.

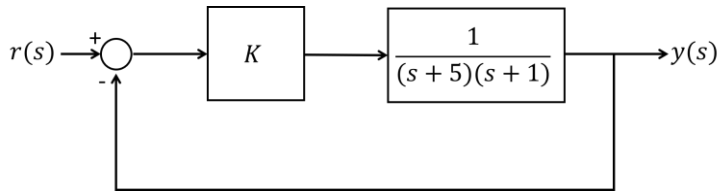


Figure 2.A – block diagram

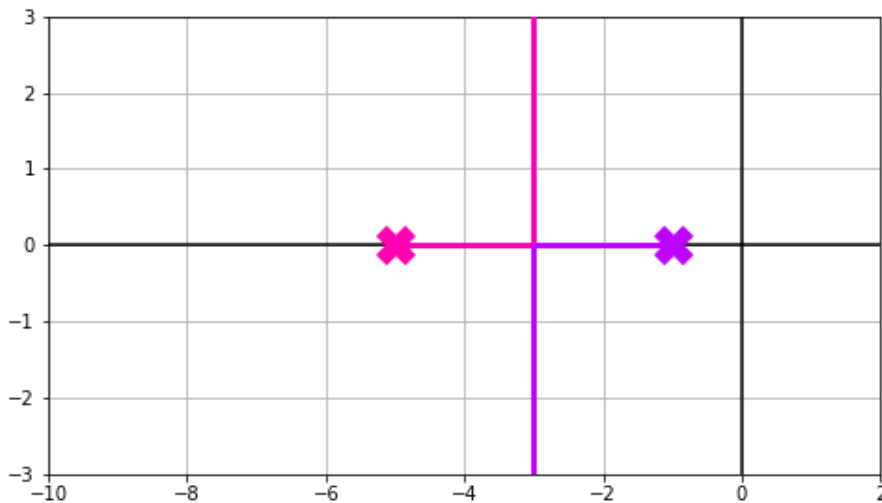


Figure 2.B – Root locus

A) [2 marks] Calculate the gain K that places the closed-loop poles at $-3 \pm j2$

Solution: The pole $-3 + j2$ is on the root locus, therefore it will be a closed-loop pole if it satisfies the magnitude criterion i.e. if the magnitude of the response is 1 for this value.

$$Q(-3 + j2) = \frac{K}{(-3 + j2 + 5)(-3 + j2 + 1)} = \frac{K}{(-2 + j2)(-2 + j2)} = \frac{K}{-8j} \therefore |G(-3 + j2)| = \frac{K}{8} \therefore K = 8$$

B) [3 marks] For the closed-loop pole locations in (A), give the (i) settling time to 5% of the final value (ii) damping ratio (iii) natural frequency (in rad/s)

Solution:

- (i) The real part of the pole is -3, meaning the time constant $\tau = \frac{1}{3}$. The 5% settling time is approximately $3\tau = 1$ second.
- (ii) The angle from the negative real axis to the pole is $\text{atan}\left(\frac{2}{3}\right) = 0.59$ radians. $\zeta = \cos(0.59) = 0.83$
- (iii) The natural frequency is the distance from the origin to the pole: $\omega_n = \sqrt{3^2 + 2^2} = 3.6$ rad/s

C) The system experiences a time delay of $T = 0.2$ seconds.

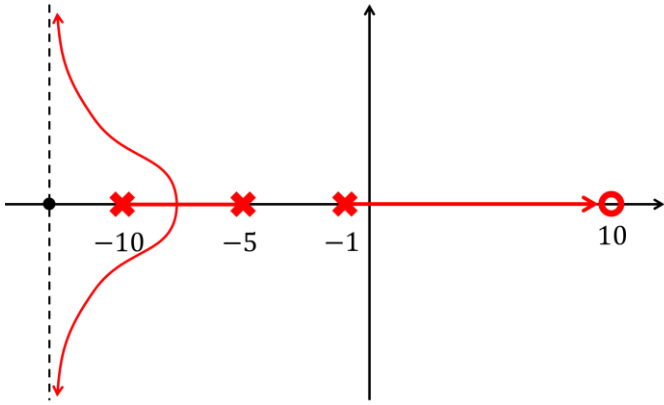
- i) [5 marks] Use a first-order Pade approximation to model the dead time, and draw the new root locus for the system. You must show the general shape and real axis segments of the loci, and, if applicable, the centroid and asymptotes.
- ii) [2 marks] Does the dead time affect the stability of the system? Explain.
- iii) [3 marks] The dead time model increases the order of the system. Would you recommend approximating the transfer function using a reduced-order model? If so, which poles are dominant, and will the model hold at all gains?

Solution:

- (i) The Pade approximation $\frac{1-0.5T}{1+0.1T} = \frac{1-0.1s}{1+0.1s}$ adds a zero at 10 and pole at -10.

The root locus exists on the real axis to the left of an odd number of critical frequencies, so it will have segments between -10 and -5, and between -1 and 10.

There are 3 poles and 1 zero, so there will be $3-1=2$ asymptotes. The angles will be $\pm \frac{\pi}{2}$ radians, and the centroid will be at $\frac{-10-1-5-(10)}{2} = -13$



- (ii) Yes: the original system was stable for all gains, while the delayed system will become unstable as the gain increases.
- (iii) The system has one pole which will always be to the right of -1, and two poles which will always be left of -5. Therefore, the leftmost poles will always be more than 5 times further from the origin than the rightmost pole, meaning the system can be approximated as first-order.

Total: 15 marks

Question 3

Decide whether each of the following statements are TRUE or FALSE. Explain your reasoning.

- A) [3 marks] The Nyquist plot in figure 3.A shows the frequency response of a system $G(s)$. This system is unstable in open loop but stable closed loop.

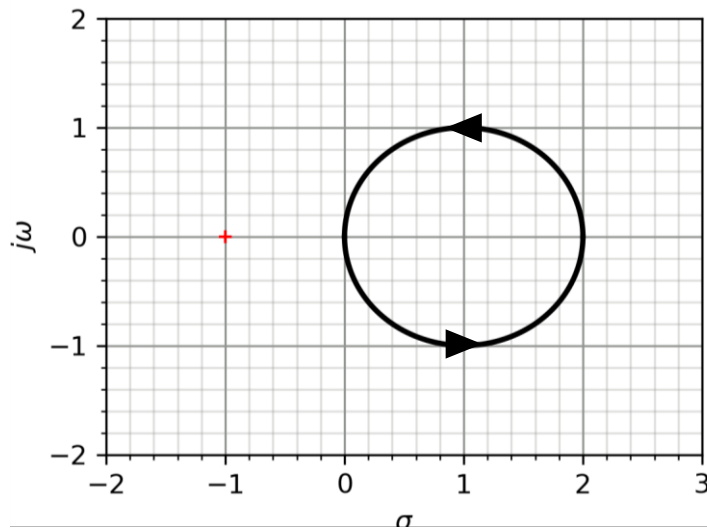


Figure 3.A Nyquist plot of $G(s)$

Solution: FALSE: The Nyquist stability criterion states $Z = N + P$, where Z is the number of unstable closed-loop poles, N is the number of clockwise encirclements around -1 and P is the number of unstable open-loop poles. If the system is unstable in open-loop, $P > 0$, but $N = 0$, so $N + P > 0$, and therefore, there must be some unstable closed-loop poles.

- B) [3 marks] The unit step response in figure 3.B was produced by the system $H(s) = \frac{1.5}{s+2}$

Solution: FALSE

For the system to have a pole at -2, it would need to be around 63% of the way to its final value at 0.5 seconds. At 0.5 seconds, it is closer to the final value than this.

Calculation of the specific parameters, which marks may also be awarded for:

The final value of the response is 1.5, so the gain of the system is 1.5.

63% of 1.5 is 0.94. The system reaches this at around 0.3 seconds, suggesting that the time constant is closer to $1/3$, so the pole is at -3.

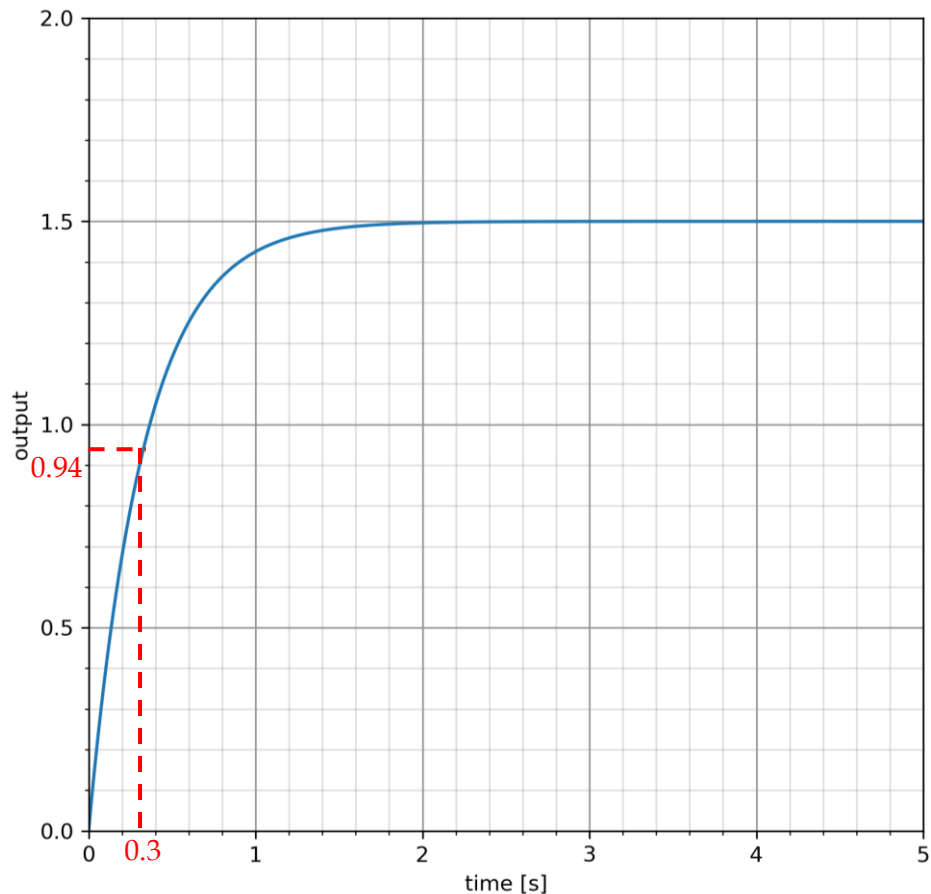


Figure 3.B unit step response

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C) [3 marks] Derivative controllers can make a system more vulnerable to noise.

Solution: TRUE: noise is typically a high-frequency signal, meaning it has a large rate of change. This will be amplified by a derivative-type controller. OR derivative controllers/ lag compensators place a zero close to the origin, which decreases the attenuation of high-frequency signals.

D) [3 marks] A controller of the form $C(s) = \frac{K}{s}$ will enable the system $\frac{1}{s+3}$ to track a step without error.

Solution: TRUE: This will create a type 1 system, which can track a first-order input without error.

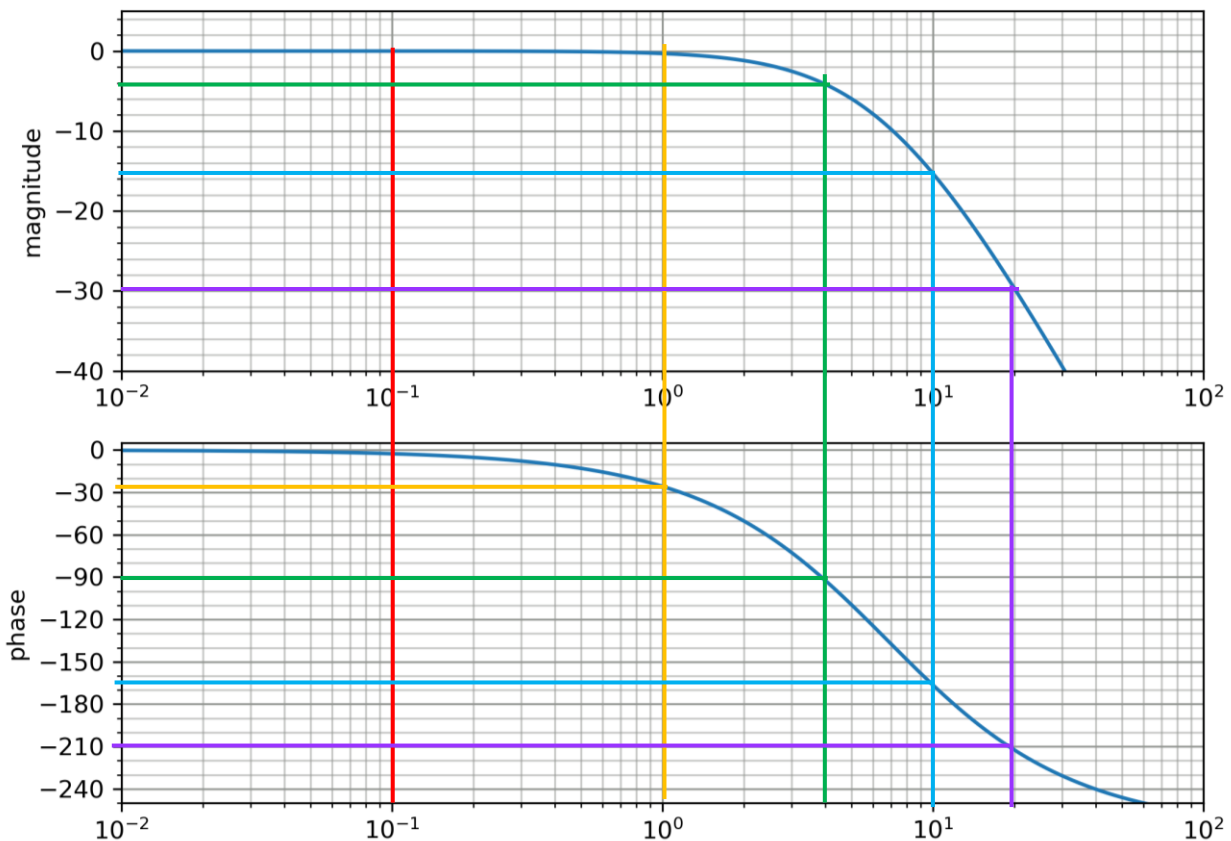
E) [3 marks] A controller of the form $C(s) = \frac{K}{s^2}$ will enable the system $\frac{1}{s+3}$ to track a ramp without error.

Hint: you may want to support your answer to part D by sketching a rough root locus.

Solution: FALSE: the controller will create a type 2 system, which should, in theory, track a second-order input without error, however, if this controller is implemented on its own, the poles at the origin will become unstable in closed loop so it will not track successfully.

Question 4

Consider the system $G(s)$ with the following open-loop frequency response.



(Construction lines added for memo)

Figure 4.A: Bode plot of $G(s)$

- A) [5 marks] Figure 4.B provides a template for an inverse Nichols chart. Sketch the frequency response shown in figure 4.A on that chart, labelling the points that correspond to the following frequencies:
 $\omega = [0.1, 1, 4, 10, 20]$

Solution: on diagram

B)

- i) [1 mark] What is the magnitude of the sensitivity function at 4 rad/s?

Solution: The point corresponding to 4 rad/s lies on the between the -1.2 and -1.4 decibel circles, so the sensitivity is around -1.3 decibels at this point. (Any answer in this range will suffice)

- ii) [2 marks] The open-loop gain is known to vary within 5% of its nominal value due to humidity. Do you expect the closed-loop gain to vary more or less than this at 4 rad/s?

Solution: The sensitivity indicates whether changes in the open loop are amplified by the closed-loop system. -1.3 decibels = 0.86, so the fluctuations will be attenuated i.e. it will show less variation.

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- C) You determine that the DC gain needs to be increased by 10 decibels to achieve sufficient tracking accuracy. Your colleagues present two different designs for feedback controllers that could be used to meet this requirement:



Godzilla

$$C_2(s) = \frac{3.16(s + 1)}{10s + 1}$$

$$C_1(s) = 3.16$$



Kong

- i) [6 marks] Sketch approximate Bode plots for each design, using the Bode plot templates in Figure 4.C. Only draw the controller, not the combined system.

Solution:

Godzilla's design is a proportional controller, so the phase plot will be the same as in Figure 4.A, while the magnitude plot is shifted up by 10:

Kong's design consists of a zero at 1 and pole at 0.1. The combined controller response can be determined by sketching the zero and pole responses, and adding them, or by evaluating the peak position and value:

Peak frequency: $\omega_p = \sqrt{(1)(0.1)} = 0.31 \text{ rad/s}$

Peak phase lag: students can use a formula from their notes, or substitute the peak frequency into the compensator equation and find the angle of the resulting complex number. Result: -55 degrees.

- ii) [6 marks] Who do you agree with? Explain why you believe their design is superior and/ or why the other design should not be used. You may support your explanation by adding sketches to your inverse Nichols chart indicating how each design will alter the response.

Solution: Kong's design should be used.

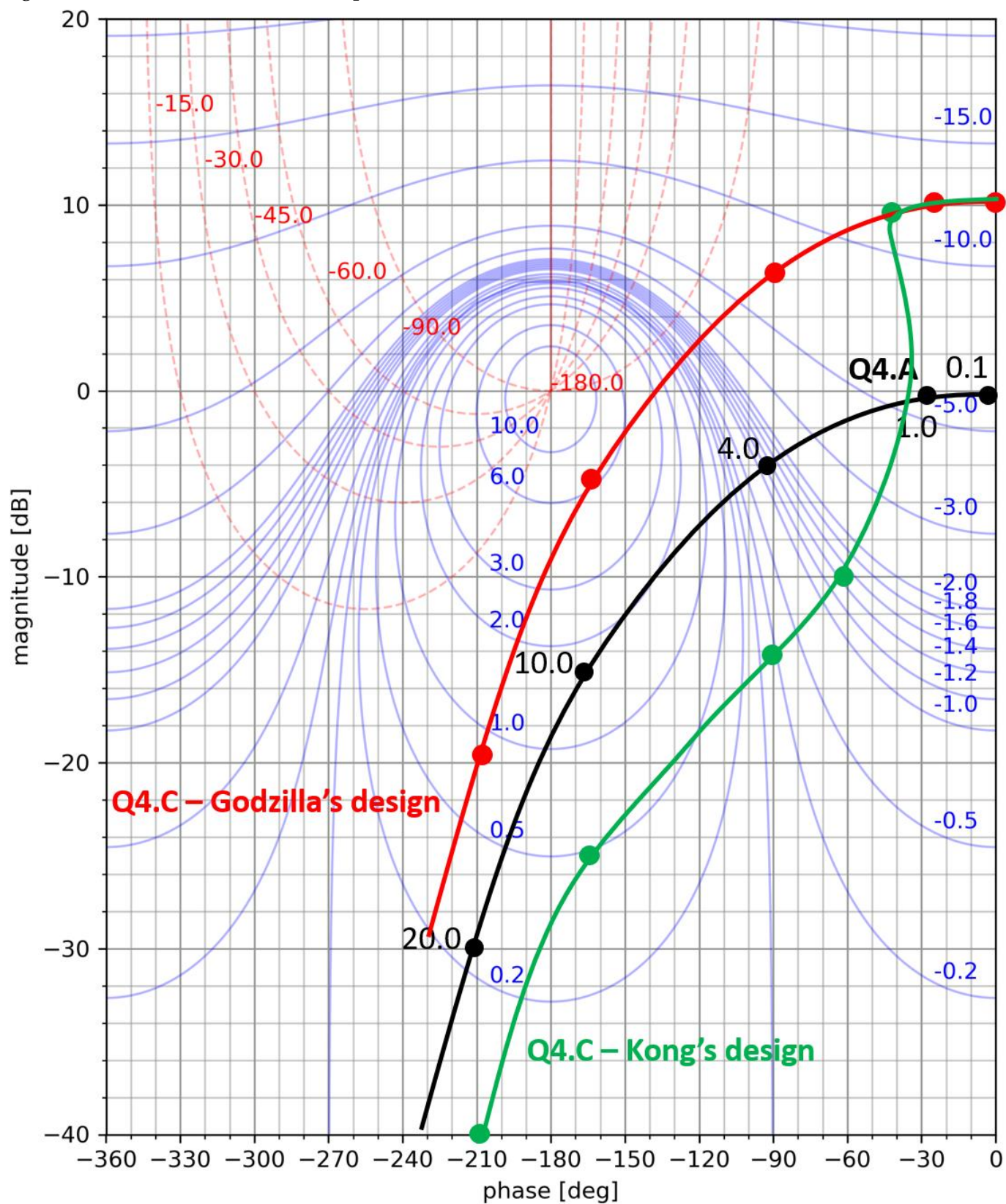
Godzilla's design increases the gain indiscriminately, so the plot will get very close to -1, i.e. the gain and phase margins will decrease, so the system will be less stable. The inverse Nichols chart shows that the response to frequencies around 10 radians per second will become very sensitive. Overall, this design makes the system less robust.

Kong's design is a lag compensator. It targets the gain increase at low frequencies, so the system's tracking accuracy is improved while still maintaining acceptable stability margins and sensitivity. These changes are indicated in Figure 4.B.

Total: 20 marks

Solution: see plot

Figure 4.B: inverse Nichols chart template



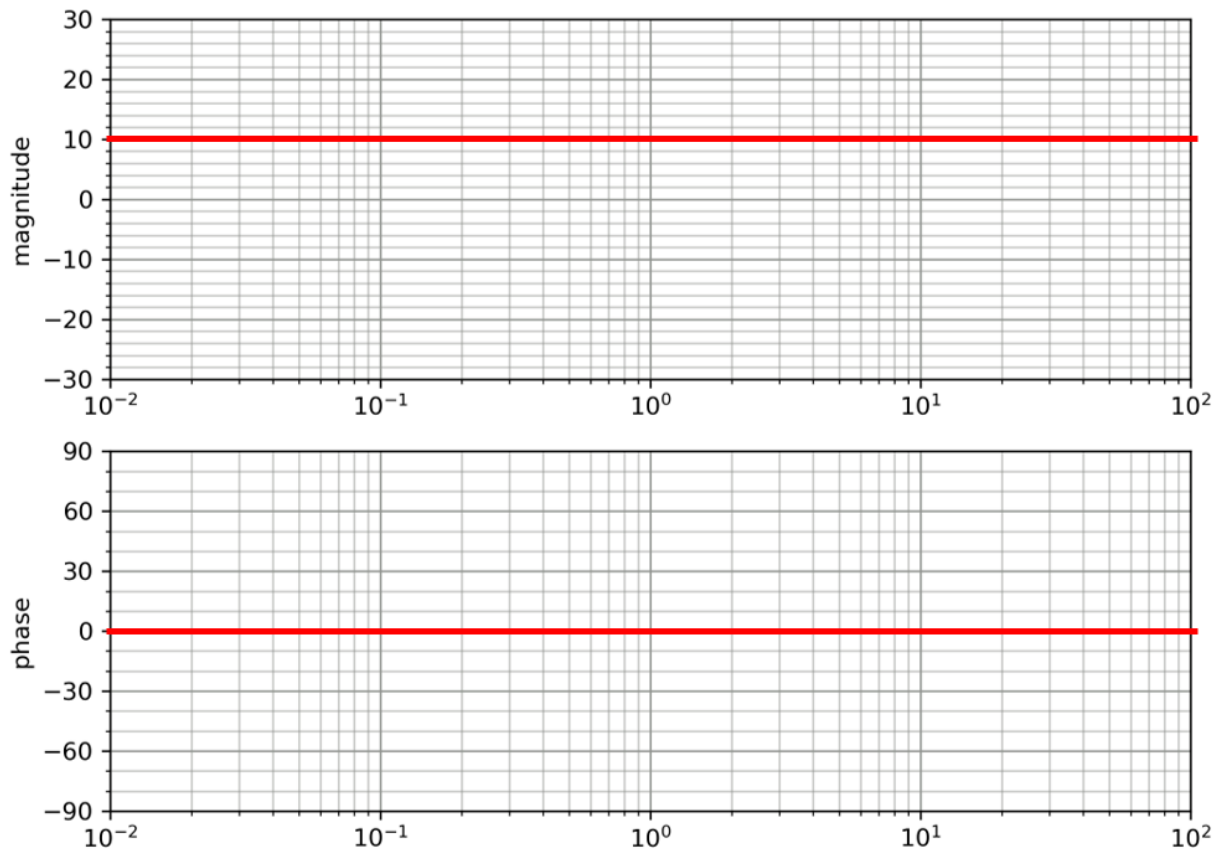


Figure 4.Ci: Godzilla's controller

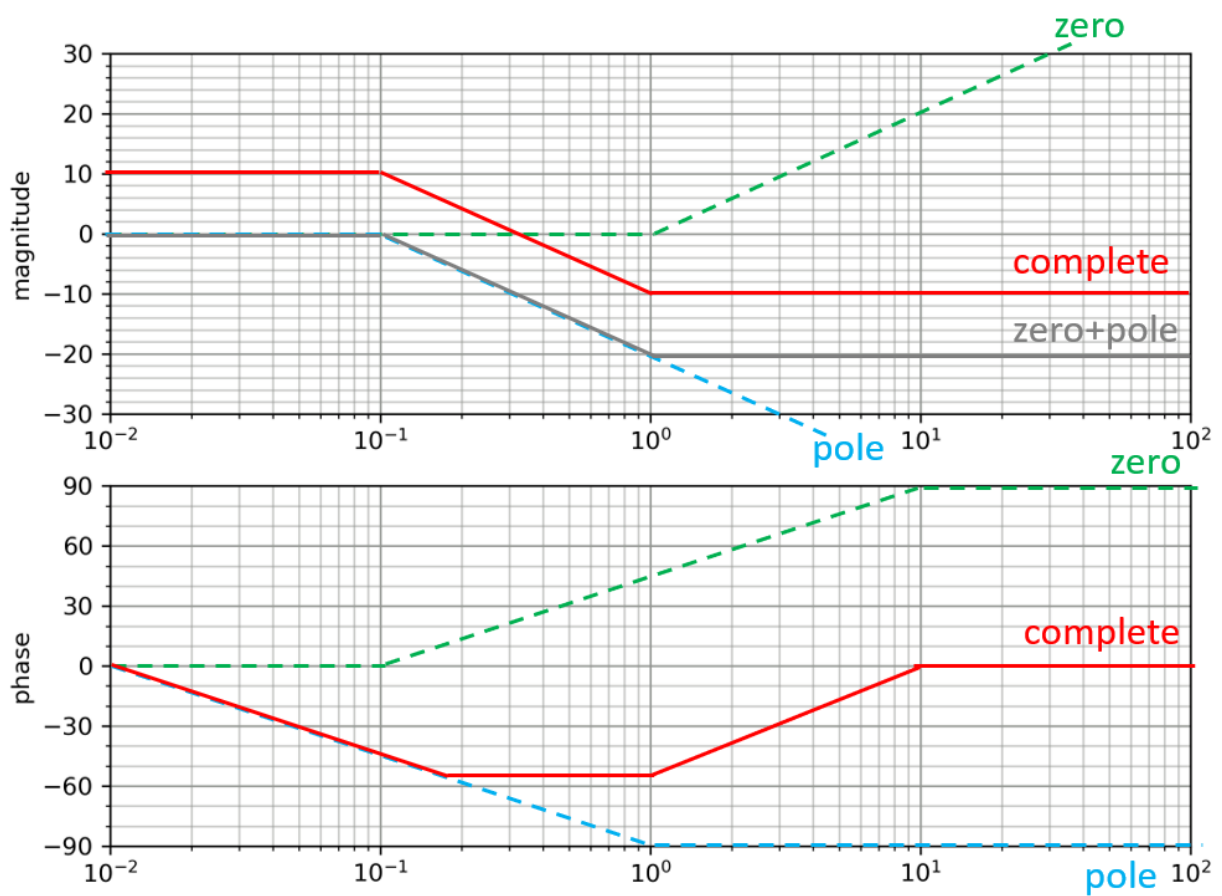


Figure 4.Cii: Kong's controller

Question 5:

Figures 5A-C depict the frequency response of a second-order system which is stable in open loop.

- A) The system is also stable in closed loop. Explain how you would know this by looking at each of the plots: (i) 5.A (ii) 5.B (iii) 5.C [3 marks]

Solution:

- (i) To check closed-loop stability on the Bode plot, we find the gain and phase margins. At the gain crossover frequency (magnitude = 1), the phase is around -150 degrees, giving a phase margin of 30 degrees. The phase does not cross -180 degrees, so the gain margin is infinite.
- (ii) The Nyquist plot does not encircle -1. Since the open-loop system is stable, there are no right-half plane poles, therefore, from the Nyquist stability criterion, the number of unstable closed-loop poles $Z = N + P = \text{zero}$.
- (iii) Stability on the Nichols chart can be determined either by finding the phase and gain margin, or in terms of crossing -1 and the Nyquist stability criterion. The margins were already determined in (i), and the response does not cross -1.

- B) [2 marks] Will the system remain stable in closed loop if you increase the gain? Explain.

Solution: yes, the gain margin is infinite.

- C) The given plots present the results of a test to evaluate the success of a controller design for the system. For each of the following performance specifications, decide whether the system meets the requirements and explain your answers based on what you observe in these tests. You may annotate the plots to support your answers.

- (i) [3 marks] Steady-state error $\leq 5\%$ for a position input

Solution: YES. the final value of the error is $\frac{1}{1+K}$ where K is the open-loop gain. $\frac{1}{1+K} \leq 0.05 \therefore K \geq 19$. Any of the given plots can be used to determine that the DC gain is 20, so it will satisfy this requirement.

- (ii) [3 marks] Damping ratio of > 0.5

Solution: NO. the damping ratio can be approximated as $\frac{A}{100} \phi_M$ where ϕ_M is the phase margin in degrees and A is the closed-loop DC gain. The phase margin is 30 degrees and the closed-loop DC gain is 0.95 so the damping ratio is around 0.285. Therefore, it does not satisfy this requirement.

- (iii) [6 marks] Settles to within 2% of final value in less than 2 seconds. (If you were unable to calculate parameters that you require for this question in previous parts, you may substitute variables or assumed values and explain how you would calculate the settling time.)

Solution: NO. The natural frequency of the system will be where it intersects the M circle corresponding to $|\frac{A}{2\zeta}|$, i.e. $20\log_{10} 0.95 - 6 - 20\log_{10} 0.285 = 4.45$ decibels. The marked frequency 1.5 rad/s is in this vicinity.

The real part of the pole is the product of the natural frequency and damping ratio: $(0.285)(1.5) = 0.43$, so the time constant is $\frac{1}{0.43} = 2.3$, which is already much larger than the required settling time.

OR: you can approximate the natural frequency as the marked frequency with the largest magnitude. The frequency 2 rad/s intersects the 10 dB circle. This leads to a slightly faster estimate of the settling time, but it is still far too slow to meet the requirement.

D) Regardless of your evaluation in B, it is decided that the performance of this controller is acceptable. It is going to be implemented digitally.

i) [1 mark] Briefly explain how sampling the system is likely to affect its frequency response.

Solution: Sampling the system will introduce phase lag that increases at higher frequencies.

ii) [2 marks] With reference to your answer in (i), would you recommend implementing the design digitally in its present form? Explain.

Solution: No: we typically want a phase margin of 45 to 60 degrees to allow for sampling effects, and this system's margin is too small. It is therefore likely to be destabilized when implemented digitally.

Total: 20 marks

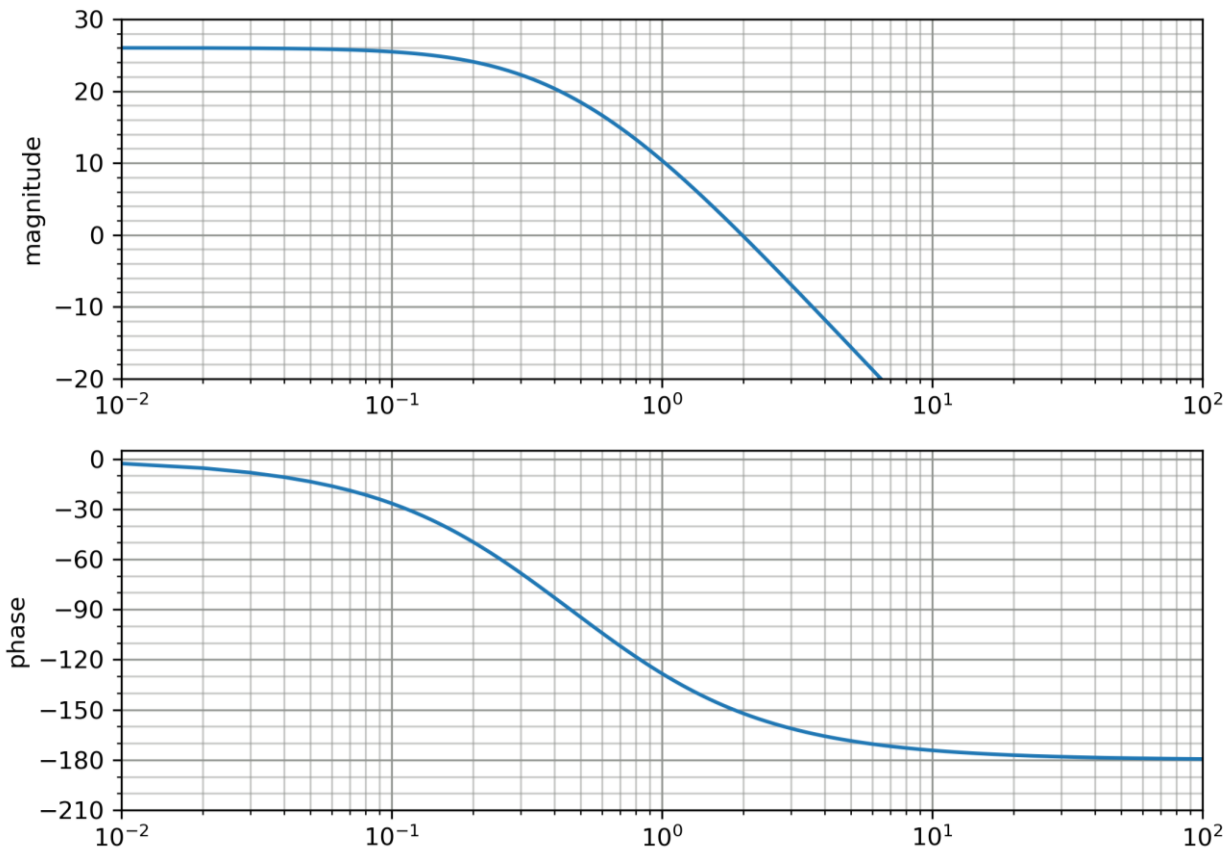


Figure 5.A: Bode plot of system

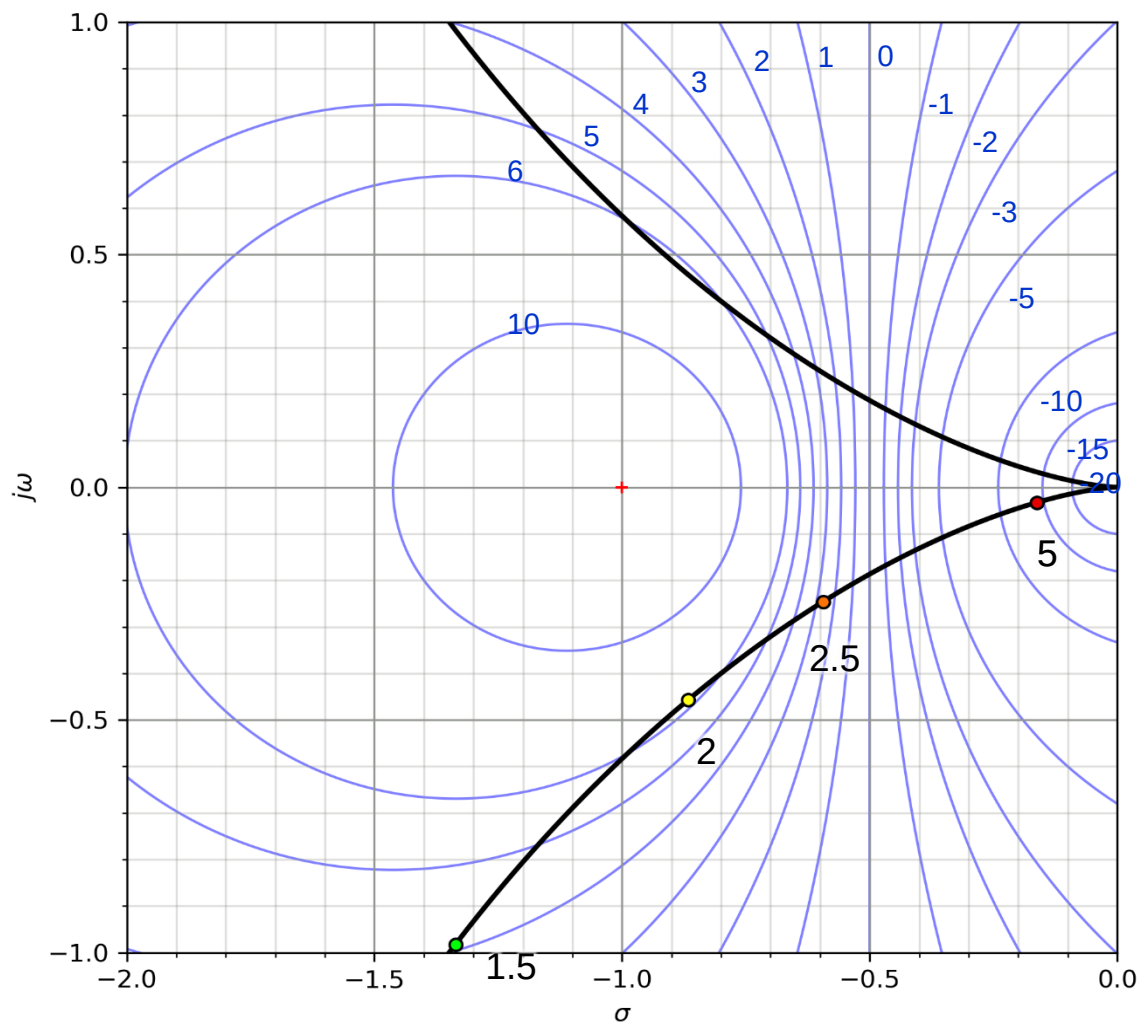
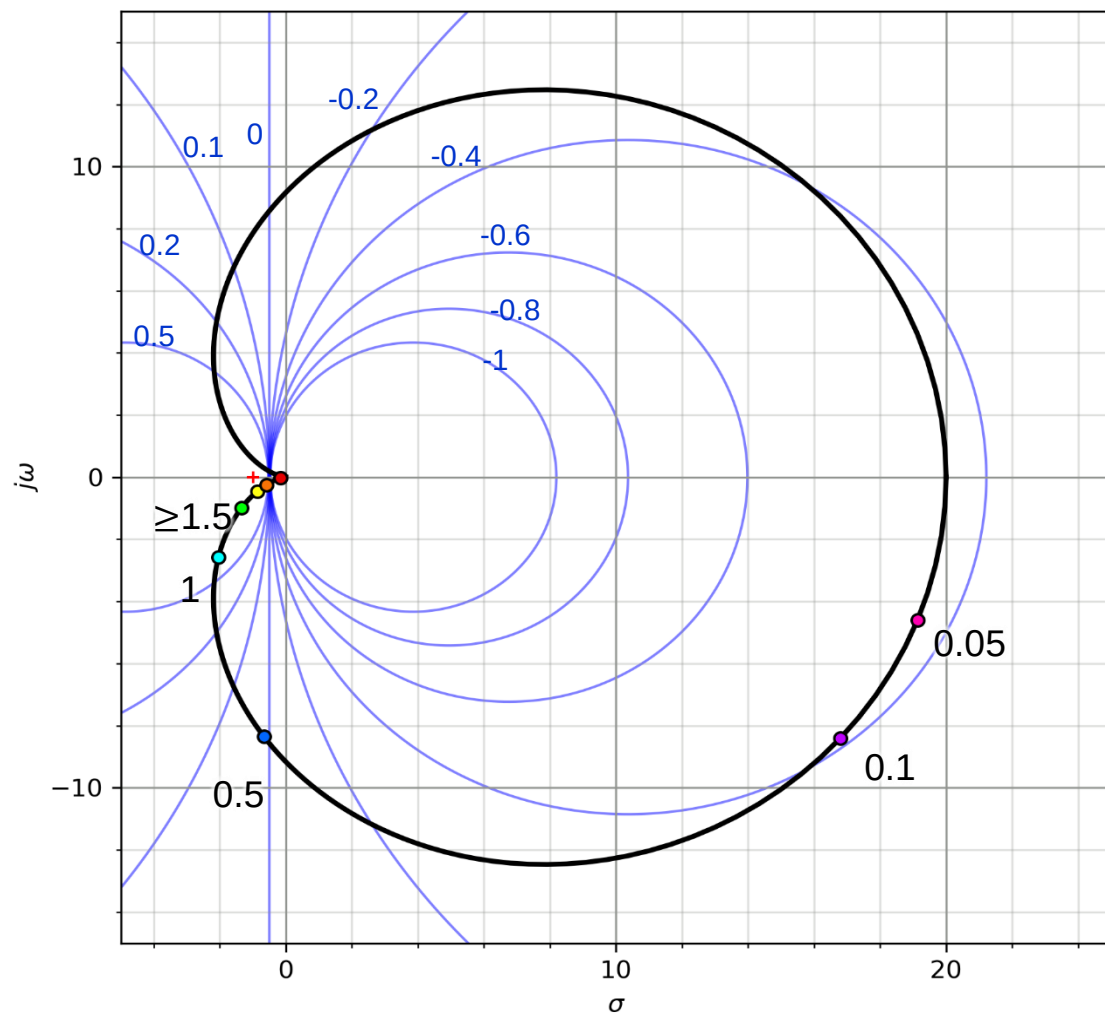


Figure 5.B Nyquist plot of system, and detailed view of Nyquist plot.

M circle magnitudes are in decibels.

Annotated frequencies are in rad/s

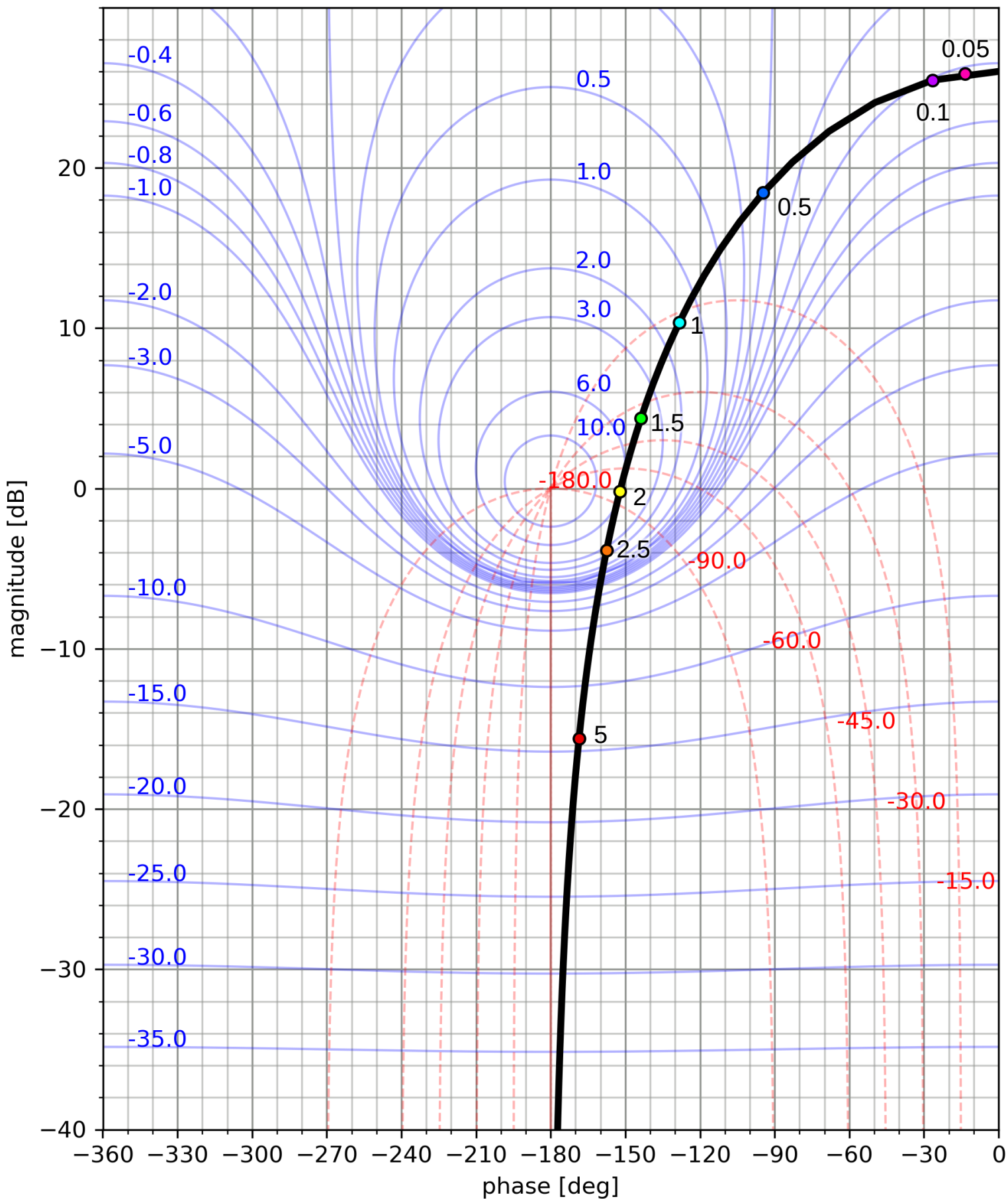


Figure 5.C Nichols chart of system. Annotated frequencies are in rad/s.

Question 6

Consider the unstable system: $G(s) = \frac{1}{(s+1)(s-1)}$

A) Theoretically, you can stabilize this system by placing it in feedback with a controller of the form

$$C(s) = K(s - z)$$

i) [1 marks] For what range of zero positions z is it possible to stabilize the system?

Solution: a zero anywhere in the left-half plane will cause the unstable open-loop pole to become stable at some gain.

ii) [2 marks] If the zero is at -0.5, find the minimum gain K for which the response will be stable.

Solution: If the zero is between the poles, the poles will always be real, so this gain can be found by substituting $s = 0$ into the combined system and solving for the K that makes the magnitude = 1:

$$1 = \frac{K(0-0.5)}{(0+1)(0-1)} \therefore K = 2$$

iii) [2 marks] Would it be possible to implement this controller in reality? Explain.

Solution: No – the transfer function is improper (more zeros than poles) so the system is not physically realizable.

B) [10 marks] Design a lead compensator to stabilize the system. The closed-loop system must satisfy the following performance requirements:

- 10% accuracy with respect to position setpoints
- Time to peak of less than 1 second
- Overshoot of less than 25%
- Phase margin of at least 60 degrees to allow for sampling

Ordinarily, a design might involve multiple iterations. If you reach the end of the process and find that it cannot satisfy the specifications based on some value you selected, you do not have to continue: you can just give a short recommendation for how the design should be modified.

Solution:

A lead compensator has the form $C(s) = \frac{K(\alpha\tau s + 1)}{(\tau s + 1)}$ where $0.1 \leq \alpha < 1$

To track a step at this accuracy, $\frac{1}{1+K} \leq 0.1 \therefore K \geq 9$

Using the %overshoot formula, you can solve to find that $\zeta \geq 0.4$.

However, the phase margin specification suggests a larger damping ratio: $\frac{0.9}{100}(60) = 0.54$

The target value should add some margin for uncertainty. Once it is chosen, you can solve for the real part of the pole using the peak time formula.

You must then calculate the angles from the existing poles to determine the angular contribution required from the compensator.

A suitable zero position is anywhere left of the target pole, up to a maximum, where the zero's angle = the compensator angle.

You should try a zero in this range, and find the corresponding pole to complete the angle criterion.

You can then find the required gain by substituting the target value into the combined frequency response and solving for the K that produces a gain of 1.

If the gain is larger than 9, the controller is suitable. If not, you must suggest that a more distant zero be selected.

Total: 15 marks